

Automating Workflow for the Calculation of Critical Properties of Percolation

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Workflow of research activity often contains a part which is repeatedly executed under a well-defined rule. Automation of such part using computer has a clear advantage over manual execution in terms of both speed and efficient use of resource. In this paper, automation of workflow using tools commonly available is discussed. As an example, we have constructed and automated a workflow for the calculation of critical properties of a two dimensional percolation model. To be specific, we focus on generating simulation data for various combinations of parameters used in the determination of its critical properties, which usually requires constant human intervention. With the automated workflow, the whole process can be completed by issuing a single command.

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I. INTRODUCTION

Research is a creative endeavor. It is search for the unknown, and thus it is hard to predict how it will unfold. However, it does not mean that whole research activities are creative. In fact, majority of them are routine, which can be described by a well-defined rule. Computer, which excels in repeatedly performing routine tasks, can be used to automate such workflows.

Workflow management system (WfMS) has been widely used in business applications for automating workflows of setting-up, executing, and monitoring pre-defined sequence of tasks [1]. The history of workflow in industry goes back to almost a century [2], and WfMS can be regarded as its evolution in the information age. Benefits of using automated workflow are well established, which include completing task in efficient and effective manner, improving internal communication, and reducing error.

Automating scientific workflow poses additional challenges. Scientific workflows are more dynamic and evolving, since research is exploratory in nature. Often, researchers carry out experiments (or simulations) in a trial-and-error manner, which requires constant change in the workflow. In addition, more sophisticated control flows are necessary in order to support large scale leading-edge simulation studies. Also, while business workflows are automated by professional software engineers, scientific workflows are maintained by scientists themselves who are experts in their domains, but not necessarily in information technology [3].

Due to these difficulties, automation of scientific workflow is mostly carried out within specific scientific domains, usually as a part of large-scale science projects. That is true for such popular workflow management systems as Kepler [4], Pegasus [5], and Triana [6]. This approach has two clear

advantages. The requirements for WfMS are much more controlled, since one has to support smaller and more homogeneous group of users. More importantly, it is possible to afford dedicated information technology personnel for the mission as a part of project.

On the other hand, majority of scientists are independent researchers who do not belong to any large scale project. They also have needs to automate their own workflows, but it is hard to find a way to accommodate them. To be fair, the WfMS projects mentioned above provide some supports to general users, but they are limited at best. Thus, it is very important to find approach to support the needs from “the rest of us”.

In this paper, I propose a way to support such needs from average scientist. It is based on shell script which is almost universally available. Furthermore, the required skills are not much different from those of average computational scientists. I choose the calculation of critical properties of two-dimensional site percolation on square lattice as example [7]. In determining critical properties of the model, one has to carry out large number of simulations for particular combinations of parameters. It requires long iterations of submitting job, checking the result, modifying parameters, and re-submitting jobs, which requires constant attention of scientists. I have automated the workflow. It consists of thousands of independent simulations (or tasks), which typically requires a couple of weeks to complete, if done manually. With the automated workflow, the whole process is executed by issuing a single command, and can be completed within a day.

II. WORKFLOW CONSTRUCTION AND AUTOMATION

1. Workflow Construction

As example, I choose two dimensional site percolation on square lattice with linear size L , where each site is randomly occupied with probability p . In the limit $L \rightarrow \infty$, the model exhibits a second order phase transition at critical point p_c [7]. The choice is made not only due to its wide applications to such diverse areas as, mathematics, physics, and material science, but also its computational aspect. The problem requires running large number of simulations for various combinations of parameters, which is quite labor intensive.

In order to measure its critical properties (for example, critical exponents), one has to simulate the model for various combinations of p and L . However, such combinations are not random. They are strategically chosen for the efficient calculation of the properties. In fact, every experienced researcher in the field has developed his/her own workflow as tacit knowledge. One of such is shown in Fig. 1. It starts with calculating spanning probability $S(p,L)$ ---probability of having a connected cluster which spans across the system---for $p \in [p_-, p_+]$ and N_s number of samples. One checks if the curve $S(p,L)$ is smooth and free of noise. If not, the number of sample N_s is increased, and the calculation is repeated, which forms the first loop. Next, one estimates critical point $p_c(L)$ from the smooth curve of $S(p,L)$, and “zooms in” around the critical point by decreasing the range $[p_-, p_+]$, and repeat the calculation. The zooming-in procedure is repeated until desired accuracy in $p_c(L)$ is achieved, which is the second loop. The whole process is repeated for other values of L , which forms the third and final loop.

If one executes the workflow by hand, one has to submit job to computer, wait for its completion, check its results, modify parameters, and re-submit job. And, the procedure is repeated hundreds or thousands times until enough data are obtained. During the whole process, scientists have to babysit the computer, which requires his/her constant attention. Also, there is a room for human error in checking

and resubmitting job. The workflow is pretty well defined, however, and there is no a priori reason why it cannot be automated.

2. Workflow Automation

The main tool chosen for the automation is shell script. It is available in any UNIX-like operating system, and it is very powerful---many functionalities have been accumulated over history, which are included in operating system itself or readily available in public domain. The whole workflow consists of two components---compiled codes and scripts which act as “glue” connecting the codes.

Let me give a detailed description on the automation process. The main code “span-2d”, which is written in FORTRAN, calculates the spanning probability $S(p, L)$ of two dimensional site percolation on a square lattice using L, p, N_s as parameters. The main computational “block” in Fig.1 is a scan over p --it calculates $S(p, L)$ for several values of p in the range of $[p_-, p_+]$ (here, 20 values of p uniformly distributed in the range are used). In order to carry out the calculations, the computer has to manage twenty batch jobs, which is done by a resource management system.

Here, I choose Swift for the management of batch jobs, which is a dataflow language for parallel computing used in cluster, cloud, and grid environments [8]. It is implicitly parallel language---all expressions may be evaluated concurrently, limited only by dataflow ordering and available processors. The main reason for the choice is its simplicity. Other resource management system such as Portable Batch System [9], HTCondor [10], and LoadLeveler [11] requires additional efforts for preparing input files, submitting jobs, and collecting the results. Using Swift, the whole calculations can be done with only a few lines of script. Furthermore, concurrent processing is built in the execution environment of Swift, and one can control the number of jobs to be executed in parallel. Here, I choose it to be 20, the number of independent jobs in the main computational block.

The first loop in Fig.1 is automated using a shell script. To be specific, one calculates $S(p,L)$, fits it with a smooth curve, and calculates fluctuation around it. If the fluctuation is significant compared to the systematic variation of $S(p,L)$, the data is decided to be too noisy. One then doubles the number of samples N_s , and repeat the procedure. Here, the curve is determined to be too noisy if the fluctuation around $S(p,L)$ is larger than 10% of its systematic variation. Such criterion works for the present case. However, it is somewhat arbitrary, and it is quite possible that one can find other criterion which works as good as, or even better than, the present one.

Automation of the second loop is also done with shell script. From “smooth” $S(p,L)$ curve, one estimates its critical point $p_c(L)$ using the criterion of $S(p_c(L),L)=1/2$. Like the first loop, it may be possible to find a better way of estimating $p_c(L)$, which can be implemented in the present method by simple change in the script. Once the critical point is found, new range of p is chosen to be the half of the previous range centered on the critical point. In other words, I want to zoom-in to the area near the critical point at each step with the magnification factor of two. The process is repeated until the range of p becomes less than the predefined accuracy.

The third and final loop is for scanning different values of L , also done with shell script. One starts with L_{min} and increase L at each cycle until it reaches the limit of L_{max} . When the whole workflow is completed, an e-mail is sent to user to notify the result. The automated workflow consists of one FORTRAN code, one Swift script, and three shell scripts. As discussed before, it is possible to replace the Swift script with shell scripts. However, that will make things more complicated---in terms of number of scripts and their length.

III. RESULTS AND DISCUSSION

1. Results

The curve $S(p,L)$ obtained from a single execution of the workflow is shown in Fig.2. Here, parameters used are $p_- = 0$, $p_+ = 1$, $N_s = 10^5$, $L_{min} = 10$, $L_{max} = 50$. Total of 1280 data points are generated (240 points for $L=10$, 260 for $L=20$, 220 for $L=30$, 300 for $L=40$, and 260 for $L=50$), not including ones discarded in the workflow. Benefits of the automation are loud and clear. If one wants to obtain the same data manually, he/she has to repeat the cycle of job submission, completion, and inspection more than thousand times. Such efforts usually take a couple of weeks, while the automated workflow can be completed within a day. In addition, scientists are freed from routine works, and can focus on more creative parts of research.

Automated workflow can help to communicate implicit knowledge [12]. One of the things scientists have to learn in carrying out research is procedures on how to get certain jobs done. For example, researchers who are studying critical phenomena have to learn exact procedures for calculating critical exponents using computer simulation. Although experienced scientists in the field know such procedures, they are in the form of tacit knowledge, which is hard to communicate. For example, teacher can “guide” student to acquire such knowledge, but it is very hard to just “teach” them. Automated workflow provides an effective way of expressing and communicating implicit knowledge, so that scientists don’t have to reinvent wheels over and over.

One other benefit of automated workflow is based on its composability, like computer program. For example, the automated workflow described above can be combined with others to form a larger workflow. Note also that workflows don’t have to be written by the same person. Thus, it is possible to build a workflow by assembling smaller workflows already available, similar to using libraries in computer programming.

2. Future Directions

An ideal workflow for the present problem should describe the entire process starting from a model to produce its critical properties with possible human intervention. What was discussed in this paper is only part of it. One obvious possibility for future works to automate other parts of the workflow such as, calculating critical exponents from the data like the one shown in Fig.2. Automation of such part was given a lower priority, since there exist little repetitive works. So, benefit from its automation is not expected to be significant. The general direction is to focus on the part of workflow which requires large amount of (human) efforts, and can be described by well-defined rules, and to let human concentrate on non-routine, creative works. One other possibility of future works is to work on other models. In particular, it may be better to focus on models exhibiting a second order transition. It is expected that the present method can be applied to such models with some modifications.

In the long run, it is desirable that most (if not, all) workflows are available as modules, and one only has to assemble them to produce workflow for the problem in hand. Indeed, this is the approach implemented successfully for visualization workflow in such examples as Advanced Visual Systems [13] and VisIt [14]. One of the big challenges in applying the winning formula to scientific workflow comes from its diversity. There are many workflow modules (or patterns) occurs in visualization process, but their number is quite small compared to that of scientific workflows. It is thus necessary to approach the problem in stages---first build workflow ecosystem for a few specific fields or problems (e.g., critical phenomena) satisfying their demands, and gradually extend them to cover entire scientific fields.

IV. CONCLUSION

In this paper, I propose an approach for automating scientific workflow for average scientist. It is based on tools commonly available and required skills are on the level of average computational scientists. As an example, I choose the calculation of critical properties of percolation, which requires large number of simulations for particular combinations of parameters. I have automated the workflow consisting of thousands of independent simulations. If done manually, the workflow requires a couple of weeks to complete. With the automated workflow, the whole process is executed by a single command, and completed in a day.

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REFERENCES

- [1] F. Leymann and D. Roller, *Production Workflow: Concepts and Techniques* (Prentice Hall, Englewood Cliffs, 1999).
- [2] F. Taylor, *The Principles of Scientific Management* (Harper & Brothers, New York, 1911).
- [3] I. Taylor, E. Deelman, D. Gannon, M. Shields, *Workflows for e-Science* (Springer-Verlag, London, 2007).
- [4] Kepler project. <https://kepler-project.org>.
- [5] Pegasus workflow management system. <https://pegasus.isi.edu>.
- [6] Triana problem solving environment. <http://www.trianacode.org>.
- [7] D. Stauffer, A. Aharony, *Introduction to Percolation Theory* (Plenum Press, New York, 1994).
- [8] Swift programming language. <http://swift-lang.org>.
- [9] Portable Batch System. <http://www.pbsworks.com>.

- [10] HTCondor Scheduler. <https://research.cs.wisc.edu/htcondor/>
- [11] LoadLeveler. <http://www.ibm.com/systems/power/software/loadleveler/>.
- [12] D. Stokes, *Pasteur's Quadrant* (Brookings Institute Press, Washington D. C., 1997).
- [13] Advanced Visual Systems. <http://www.avs.com/>.
- [14] VisIt. <https://wci.llnl.gov/simulation/computer-codes/visit/>.

Figure Captions.

Fig. 1. Workflow for the calculation of critical properties of percolation. It starts with a set of initial parameters, generates data with three loops of iterations, and ends by sending a notification to user.

Fig. 2. Calculated $S(p,L)$ from a single execution of the automated workflow, which took less than a day. It contains total of 1280 data points, not including ones discarded.

Fig. 1.

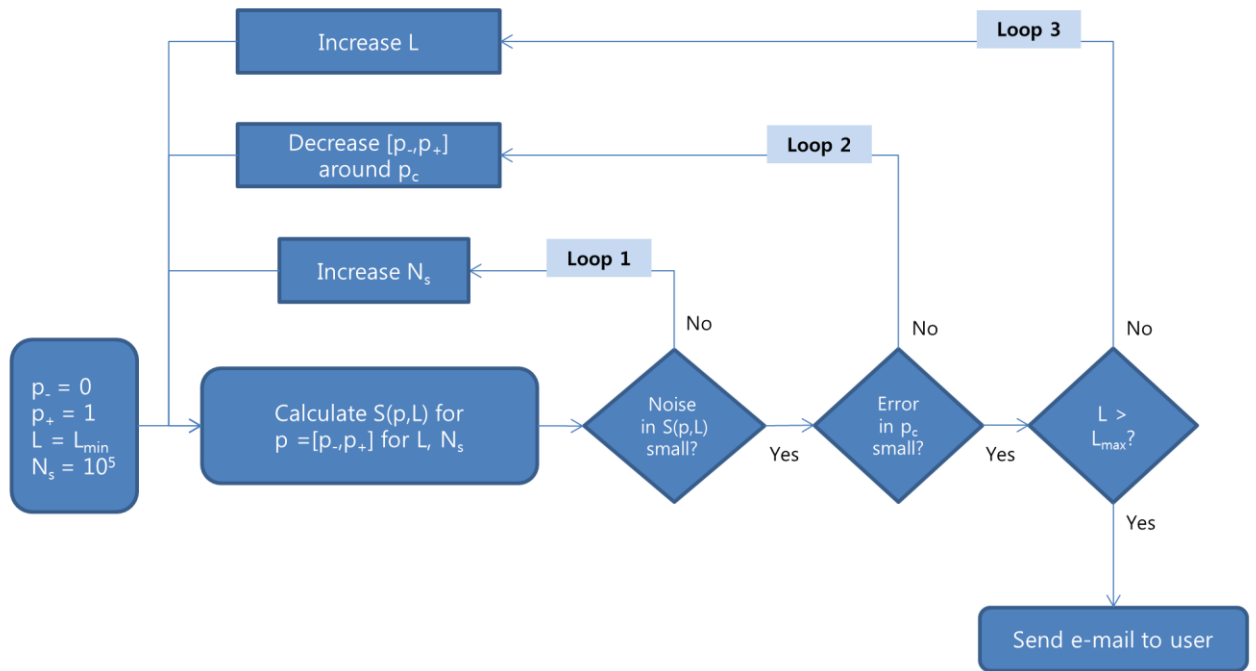


Fig. 2.

